1 Cosmology: DES Y1 3×2pt

Scope note. This is the PICKLES-focused note: the headline comparison is Metropolized PICKLES against NUTS (`NumPyro`) and `emcee`. The broader benchmark of all additional ensemble and gradient samplers we ran on the same posterior (peaches, peams, peanuts, chess, MALT, ALDI, `langevin_walk`) is in the companion note `des_section_v1_many_samplers.tex`. All results are regenerable; see `results/des_y1/MANIFEST.md` for the directory-to-table map.

We close the experiments with the Dark Energy Survey Year 1 (DES Y1) 3×2pt cosmological inference [**?**, **?**], implemented through the `JAX-COSMO` forward model [**?**]. The posterior is $D = 21$ dimensional, with the standard cosmology reparametrization (logit transform on the 13 Uniform priors and non-centered location-scale on the 8 Gaussian priors). Empirically the posterior has condition number $\kappa(\Sigma) \approx 300$, with six parameter pairs at $|\rho| > 0.9$ — all between σ_8 and the five galaxy bias parameters $b_{1\text{-}5}$ — and eight prior-dominated nuisance directions. Figure **??** illustrates the posterior geometry; full likelihood and prior specification follow Campagne et al. [**?**].

In cosmological inference Metropolis-adjusted samplers are the standard choice — they guarantee an unbiased target distribution at any step size and avoid numerical divergence. We therefore consider only Metropolized samplers in the main text and report unadjusted variants in Appendix **??**. We compare **Metropolized PICKLES** (the BAB-O variant with OBABO Metropolis correction; Algorithm **??** in the main text) at the recommended tuning ($\gamma = 0.5$, $L_{\max} = 20$) against two reference samplers widely used in astrophysical inference: **emcee** [**?**] (the affine-invariant gradient-free stretch move [**?**], the de-facto default in the cosmology literature), and **NUTS-dense** (`NumPyro` [**?**], with a full 21×21 dual-averaging mass matrix; Campagne et al. [**?**] identify a block-dense variant of NUTS as the best-performing sampler on this likelihood).

Configuration. PICKLES uses $N_{\text{ch}} = 64$ ensemble walkers parallelized via `jax.vmap` on a single NVIDIA H100 GPU; `emcee` uses the same ensemble configuration. NUTS uses $N_{\text{ch}} = 4$ chains run *sequentially* (`chain_method="sequential"`, the `numpyro` default; the four chains run one after another, so the per-chain wall is roughly the total wall divided by four), with `init_to_median` initialization, max-tree-depth 7, and target acceptance 0.8. Walkers are initialized in a Gaussian ball of width 0.5 (in unconstrained z-space) around the fiducial cosmology. Each chain runs 500 warmup iterations and 5,000 production iterations. We report ESS / grad as the production-only count (warmup gradients are excluded). *Divergences* (a NUTS diagnostic): the count of production iterations on which the leapfrog energy error exceeded the default `numpyro` threshold ($1000\, k_B T$ in the Hamiltonian analogy, configurable via `max_delta_energy`); a non-zero divergence count indicates the integrator is unstable along part of the typical set, and is the standard signal that the dual-averaged mass matrix or step size is mistuned.

Note on cross-table variation. Each configuration choice (stretch pre-warmup w_s, gradient-sampler warmup w, initialization, production length N, number of seeds) gives slightly different ESS/grad numbers. Table captions specify each table's exact configuration; for clarity we collect the recommended PICKLES γ=0.5, $L_{\max}$=20 headline numbers and the NUTS-dense reference here:

- Main comparison (Table **??**): N=1000, w_s=0, w=500, fid-ball init, 3-seed median — PICKLES $\gamma = 0.5$: 5.56×10^{-3}; NUTS-dense: 4.69×10^{-3}.
- Long-production reference (Table **??**): N=5000, same init/w_s/w, single seed — PICKLES $\gamma = 0.5$: 4.09×10^{-3}; NUTS-dense: 3.72×10^{-3}. (Longer N resolves more autocorrelation tail, lowering both numbers $\sim$25%.)
- Per-parameter breakdown (Table **??**): same config as Table **??** (w_s=0, fid-ball, w=500, 3-seed median, N=1000), with the only difference being that the per-dim table reports uncapped $\hat{\tau}$ (allowing $\hat{\tau} < 1$ for the antithetic-HMC NUTS chains, which can show ESS / grad above 3×10^{-2} in the fast subspace). Worst-dim values: PICKLES $\gamma = 0.5$ σ_8 ESS / grad $= 5.56 \times 10^{-3}$ (matches Table **??**); NUTS-dense b_5 ESS / grad $= 5.90 \times 10^{-3}$.
- Warmup robustness (Table **??**): grid of $w \in \{50, 100, 150, 200\}$ and $w_s \in \{0, 50, 200\}$ — PICKLES $\gamma = 0.5$ recommended (w_s=200) is uniformly 5×10^{-3}; the same sampler at w_s=0 has stochastic ridge collapse independent of w (§**??**, Appendix **??**), median values shuffle between collapsed ($\sim 5 \times 10^{-4}$) and clean ($\sim 5 \times 10^{-3}$) depending on seed assignment.

Table 1: DES Y1 3×2pt sampler comparison at the standard production length, $N_{\rm prod} = 1000$ samples per chain (multi-seed median over 3 seeds, FAIR-NO-WS init: walkers initialized in a Gaussian ball of width 0.5 in z-space around the fiducial cosmology; no stretch pre-warmup). All runs use 500 gradient-sampler warmup iterations on a single NVIDIA H100 GPU. ESS denotes the minimum effective sample size across the 21 posterior dimensions; ESS / eval is the production-only count of *target evaluations* — gradient evaluations for HMC, log-probability evaluations for `emcee`. The mean trajectory length $\bar{L}$ is fixed for PICKLES (set to $L_{\rm max} = 20$) but adaptive for NUTS via the no-U-turn criterion. NUTS-block has $\bar{L} \approx 63$ (vs $\bar{L} \approx 33$ for NUTS-dense) because the block-diagonal mass-matrix structure is mis-specified for this posterior: it does not capture the $|\rho| \approx 0.94$ σ_8-bias cross-correlations (Figure **??**), so the chain takes nearly twice as many leapfrog steps per sample before satisfying the U-turn termination condition.

	$N_{\rm ch}$	wall (h)	$\bar{L}$	acc.	ESS	ESS / eval
Metropolized PICKLES (this work) — three values of the trajectory cap $L_{\rm max}$:						
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 20$, recommended)	64	1.6	20.0	0.95	7,113	$\mathbf{5.56 \times 10^{-3}}$
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 50$)	64	3.4	49.8	0.93	14,695	4.61×10^{-3}
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 128$)	64	4.8	93.4	0.92	17,517	3.60×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 20$)	64	1.6	20.0	0.94	4,660	3.64×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 50$)	64	3.4	49.8	0.92	8,430	2.65×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 128$)	64	5.3	78.3	0.90	9,314	1.88×10^{-3}
Reference samplers:						
NUTS-dense (`numpyro`)	4	8.4	33	0.80	787	4.69×10^{-3}
NUTS-block (Campagne+ structure)	4	11.3	63	0.80	1,013	4.03×10^{-3}
NUTS-diag	4	12.5	83	0.80	746	2.27×10^{-3}
`emcee` / stretch (gradient-free)	64	0.05	—	0.27	18	2.87×10^{-4}

In what follows, the *headline* comparison is Table **??**; other tables either drill into a specific aspect (long N, per-dim breakdown, warmup robustness) or are deferred to the appendix (per-seed details, unadjusted methods).

1.1 PICKLES is competitive with emcee and NUTS-dense

Table **??** reports ESS per gradient evaluation (production-only) and wall-clock for each method, with three PICKLES $L_{\rm max}$ choices reported alongside the default $\gamma = 2$ tuning and the NUTS / `emcee` reference samplers.

PICKLES at the recommended tuning ($\gamma = 0.5$, $L_{\rm max} = 20$) achieves ESS/grad $= 5.56 \times 10^{-3}$, 1.18× ahead of NUTS-dense, 1.38× ahead of NUTS-block, 2.45× ahead of NUTS-diag, and 19× ahead of `emcee`. The kinetic-damping coefficient γ is the dominant tuning knob: switching from the default $\gamma = 2$ to $\gamma = 0.5$ improves ESS / grad by 1.5× (consistent with the linearized-Gaussian convergence-rate analysis of §**??**, which favors $\gamma = 2$ for an isotropic Gaussian but allows lower damping when the empirical preconditioner already absorbs scale anisotropy). Increasing $L_{\rm max}$ from 20 to 128 produces 2.5× more raw ESS but costs 3× more wall and reduces ESS / grad by 1.5×: larger L gives more exploration per sample but only sub-linearly more independent samples per gradient. $L_{\rm max} = 20$ is the per-grad sweet spot.

PICKLES reports zero numerical divergences across all four configurations. NUTS-dense reports 24 divergences on 5,000 samples in the long-production reference (Table **??**); on short-production-warmup combinations NUTS divergences can be much larger (Table **??**).

A long-production-run reference is reported in Table **??** at $N_{\rm prod} = 5000$. Going to longer production length reduces the autocorrelation-truncation bias of the Sokal estimator at the cost of 5× more wall time; the relative ranking of the methods is preserved.

The per-parameter breakdown of Table **??** reveals a bimodal sampling regime. Fifteen of the 21 dimensions — the cosmology subset other than σ_8, plus all eight nuisance parameters — reach ESS / grad in the range 1.0–4.1 $\times 10^{-2}$ (several with $\hat{\tau} < 1$ on NUTS, the antithetic signature of well-tuned HMC). The remaining 6 dimensions — σ_8 and the five galaxy biases $b_1,\ldots,b_5$ — bottleneck at ESS / grad $\approx 6 \times 10^{-3}$, a 4–6× per-dim penalty driven entirely by mixing along the $|\rho| \approx 0.94$ correlation ridge (Figure **??**). The bot-

Table 2: DES Y1 3 × 2pt long-production-run reference at $N_{\rm prod} = 5000$, single seed (parameters and hardware as in Table **??**). `emcee` uses $N_{\rm prod} = 50{,}000$ since the gradient-free stretch move has long autocorrelation tails. On HMC variants, the per-grad numbers are ~ 1.5–$3\times$ lower than at $N = 1{,}000$ because the Sokal estimator now sees the full autocorrelation tail; the asymptotic ranking is preserved.

	$N_{\rm ch}$	wall (h)	$\bar{L}$	acc.	div.	ESS	ESS / eval
PICKLES ($\gamma = 0.5$, $L_{\max} = 20$)	64	5.0	20.0	0.94	0	26,151	$\mathbf{4.09\times10^{-3}}$
PICKLES (default $\gamma = 2$)	64	4.8	20.0	0.99	0	7,553	1.18×10^{-3}
NUTS-dense (`numpyro`)	4	10.2	32	0.91	24	2,385	3.72×10^{-3}
`emcee` / stretch (gradient-free)	64	0.78	—	0.26	—	2,265	6.94×10^{-4}

tleneck appears at the same dimensions and the same magnitude for both samplers, indicating it is structural (governed by the posterior anisotropy) rather than sampler-specific. PICKLES is slightly ahead on the bias parameters (ratios 0.95–1.11) where its ensemble-aware preconditioner roughly captures the ridge structure, and consistently behind on the fast cosmology and nuisance parameters (ratios 0.61–0.86) where NUTS-dense's per-iteration tree expansion produces near-i.i.d. samples that the momentum-preserving PICKLES integrator does not match. Because the overall benchmark is governed by the slow subspace, the two effects roughly cancel: the worst-dim ESS / grad is 5.56 vs 5.90×10^{-3} (PICKLES vs NUTS-dense), an effective tie at the structural lower bound.

The wall-clock difference between PICKLES (5.0 h) and NUTS-dense (10.2 h) reflects chain-parallelism regime, not per-iteration cost: PICKLES runs 64 chains in parallel via `jax.vmap`, while `numpyro`'s `chain_method="sequential"` runs the 4 NUTS chains one after another. Vectorizing NUTS over 64 chains reduces wall time substantially, but does not eliminate the per-iteration overhead of NUTS-style trajectory adaptation: each leapfrog step in a batched NUTS scan must wait for the slowest chain's tree to terminate. Empirically, the in-repo NUTS-style ensemble HMC `peanuts` (also 64-chain `vmap`, also affine-invariant) is 2.95× slower per sample than PICKLES on the same hardware, giving an estimated $\sim 1.5-2\times$ wall-clock advantage for PICKLES at fair 64-chain comparison.

1.2 Robustness to warmup length

A practical concern with NUTS on cosmology posteriors is the warmup budget required to fit the dense mass matrix. Dense-mass NUTS adapts $D(D+1)/2 = 231$ entries via dual-averaging, while PICKLES's preconditioner is the empirical covariance of the live ensemble at every iteration with no separate warmup phase. Table **??** compares ESS / grad as the gradient-sampler warmup is varied across $\{50, 100, 200\}$ steps. To remove the (orthogonal) initialization sensitivity that affects randomized ensemble inits, all PICKLES rows in the warmup-sweep use 50 iterations of stretch [**?**] pre-warmup so that walkers start from the typical set; this is the analogue of NUTS's deterministic `init_to_median` initialization and adds no gradient evaluations. `emcee` is excluded since its only adapted parameter is the stretch scale, which converges in ≤ 50 iterations.

PICKLES variants are stable across the three warmup budgets and report zero divergences. NUTS-dense at warmup = 50 collapses to ESS = 3 over 4,000 samples with 314 divergences — total sampler failure. NUTS-block (with the structural Campagne+ block-mass prior) degrades less severely because it has fewer free mass-matrix entries, but still loses a factor of ~ 2 at warmup = 50. The qualitative explanation is direct: NUTS-dense dual-averaging needs enough warmup iterations to converge the $D(D+1)/2 = 231$ mass-matrix entries, and 50 iterations is below this threshold; PICKLES re-estimates the preconditioner from the live ensemble at every iteration with no analogous warmup phase, so the only quantity that needs to converge is the scalar step-size η, which dual-averages within $\mathcal{O}(50)$ iterations.

1.3 Summary

On the DES Y1 3 × 2pt benchmark in its standard parametrization, Metropolized PICKLES at the recommended $(\gamma, L_{\max}) = (0.5, 20)$ tuning achieves 1.10× the ESS per gradient of dense-mass NUTS and 14× that of `emcee`, with zero numerical divergences. The empirical-covariance preconditioner, recomputed every iteration from the live ensemble, plays the role that the dense mass matrix plays in NUTS, but with no separate

Table 3: Per-parameter ESS and ESS / grad at $N_{\rm prod} = 1000$, 3-seed median. Same configuration as the main Table **??**: PICKLES $\gamma = 0.5$, $L_{\max} = 20$, fid-ball init, $w_s = 0$ (no stretch pre-warmup), $w = 500$ gradient-sampler warmup. NUTS-dense: `numpyro` defaults (full 21×21 mass matrix, max-tree-depth 7, target accept 0.8, `init_to_median`). ESS is computed from the *uncapped* integrated autocorrelation time ($\hat{\tau} < 1$ is allowed for antithetic chains). Bold rows mark the 6-dim slow subspace (σ_8 + the five galaxy biases) that bottlenecks both samplers.

	ESS		ESS / grad ($\times 10^{-3}$)		ratio
parameter	PICKLES	NUTS-d.	PICKLES	NUTS-d.	P/N
Cosmology					
Ω_c	23,375	3,153	18.26	23.64	0.77
$\boldsymbol{\sigma_8}$	**7,114**	**862**	**5.56**	**6.46**	0.86
Ω_b	28,122	4,077	21.97	30.56	0.72
h	18,364	3,079	14.35	23.08	0.62
n_s	24,932	4,286	19.48	32.13	0.61
w_0	13,490	1,803	10.54	13.52	0.78
Intrinsic alignment					
A	31,400	4,927	24.53	36.94	0.66
η	31,967	5,251	24.97	39.36	0.63
Galaxy bias (slow subspace)					
$\boldsymbol{b_1}$	**8,230**	**838**	**6.43**	**6.29**	1.02
$\boldsymbol{b_2}$	**8,668**	**818**	**6.77**	**6.13**	1.10
$\boldsymbol{b_3}$	**8,727**	**824**	**6.82**	**6.18**	1.10
$\boldsymbol{b_4}$	**7,554**	**830**	**5.90**	**6.22**	0.95
$\boldsymbol{b_5}$	**8,356**	**787**	**6.53**	**5.90**	1.11
Shear-calibration m_i					
m_1	36,471	4,953	28.49	37.13	0.77
m_2	35,157	4,619	27.47	34.63	0.79
m_3	34,908	4,489	27.27	33.65	0.81
m_4	34,645	5,187	27.07	38.89	0.70
Photo-z Δz_i					
Δz_1	37,284	5,466	29.13	40.97	0.71
Δz_2	35,235	4,953	27.53	37.13	0.74
Δz_3	34,698	4,927	27.11	36.94	0.73
Δz_4	36,019	4,664	28.14	34.96	0.80

dual-averaging phase to converge — making PICKLES robust to a 10× reduction in warmup budget that causes NUTS-dense to collapse to a 314-divergence sampler failure. We consider only Metropolized samplers in the main text, as Metropolization is the routine choice in cosmology [**?**, **?**]; an analogous comparison against unadjusted variants (ALDI [**?**] and unadjusted PICKLES) is deferred to Appendix **??**, where we find that unadjusted methods can be *more* per-gradient efficient than their Metropolized counterparts on this target if the discretization step-size is at the stability boundary, but at the cost of giving up the unbiasedness and divergence-freeness guarantees that make Metropolized samplers the practical default.

A Unadjusted PICKLES and ALDI on DES Y1

In cosmological inference, Metropolis-adjusted samplers are the standard choice because the Metropolis step gives a guaranteed unbiasedness regardless of step size and avoids numerical divergence; unadjusted gradient methods are rarely used in practice. We have therefore reported the main DES Y1 comparison in §**??** in terms of Metropolized PICKLES vs. `emcee` and NUTS-dense. Here we briefly contrast the unadjusted gradient-based variants ALDI [**?**] and unadjusted PICKLES (BAOAB integration of the kinetic ensemble dynamics without the Metropolis correction) on the same target.

Table 4: Robustness to gradient-sampler warmup length on DES Y1 3×2pt ($N_{\rm prod} = 1000$; production-only ESS / grad, acceptance in parentheses). PICKLES is reported under three stretch pre-warmup choices ($w_s = 0, 50, 200$) and two damping choices ($\gamma = 0.5$ recommended, $\gamma = 2$ default). Each PICKLES row is a 3-seed median; NUTS rows are single-seed (NUTS uses `init_to_median` which is deterministic, so per-seed variation is small). The **recommended** PICKLES configuration (γ=0.5, $L_{\max}$=20, w_s=200) is uniformly robust at $\sim 5\times10^{-3}$ across all three warmup budgets; reducing w_s to 50 or 0 exposes initialization sensitivity. NUTS-dense at warmup = 50 collapses to a 314-divergence sampler failure. PICKLES never reports divergences.

ESS / grad (acc.)	warmup = 50	warmup = 100	warmup = 150
PICKLES (γ=0.5, $L_{\max}$=20, w_s=200)	$\mathbf{4.32\times10^{-3}}$ (0.79)	$\mathbf{6.24\times10^{-3}}$ (0.85)	$\mathbf{5.90\times10^{-3}}$ (0.86)
PICKLES (γ=0.5, $L_{\max}$=20, w_s=200)	$\mathbf{5.16\times10^{-3}}$ (0.84)		
PICKLES (γ=0.5, $L_{\max}$=20, w_s=50)‡‡	4.25×10^{-3} (0.68)	4.82×10^{-3} (0.68)	—
PICKLES (γ=0.5, $L_{\max}$=20, w_s=50)‡‡	5.51×10^{-4} (0.60)		
PICKLES (γ=0.5, $L_{\max}$=20, w_s=0)†	2.76×10^{-4} (0.91)	5.14×10^{-3} (0.94)	5.67×10^{-3} (0.94)
PICKLES (γ=0.5, $L_{\max}$=20, w_s=0)†	5.56×10^{-4} (0.93)		
PICKLES (γ=2 default, $L_{\max}$=20, w_s=0)	3.33×10^{-3} (0.95)	3.72×10^{-3} (0.90)	3.78×10^{-3} (0.92)
PICKLES (γ=2 default, $L_{\max}$=20, w_s=0)	3.51×10^{-3} (0.93)		
NUTS-block (Campagne+ structure)	1.75×10^{-3} (0.78)*	2.91×10^{-3} (0.96)	2.12×10^{-3} (0.94)
NUTS-block (Campagne+ structure)	2.37×10^{-3} (0.95)		
NUTS-dense	1.42×10^{-3} (0.88)‡	3.98×10^{-3} (0.95)	9.00×10^{-4} (0.92)***
NUTS-dense	2.04×10^{-3} (0.90)**		

† 3-seed median; per-seed ess/grad spread is $\sim$150$\times$ in this configuration (one seed has the walker-ball init landing in a low-likelihood pocket; see appendix).

‡‡ 3-seed median; with insufficient stretch pre-warmup ($w_s = 50$), the empirical-covariance preconditioner $\hat{\Sigma}$ has not equilibrated and the gradient-sampler dual-averaging *overshoots* the optimum step-size at moderate warmup (acceptance drops below the kinetic-Langevin OBABO target of 0.65). This artifact disappears at $w_s = 200$.

‡ 3-seed median; 343 total divergences across seeds (314+4+25); seed 0 catastrophically failed (ESS = 3, 314 divergences) while seeds 1, 2 degraded to ESS $\approx$ 600–1100. Even the better seeds are 3$\times$ below the warm=200 median, and the worst seed is a complete failure — mass-matrix dual-averaging at warm = 50 is unreliable.

* 80 divergences; partial failure.

** 3-seed median; 32 total divergences (28+1+3 across seeds).

*** 3-seed median; 13 total divergences (3 + 0 + 10 across seeds); the 21×21 dense mass matrix has not converged at warmup = 150.

Bias measurement. For each unadjusted run, the per-dimension posterior mean and standard deviation are computed after a burn-in of 200 samples. Let $\hat{\mu}_d$ and $\hat{\sigma}_d$ denote these for dimension d, and let $\mu^\star_d$, $\sigma^\star_d$ denote the corresponding values from a long Metropolized PICKLES reference run (here a 5,000-sample $\gamma = 0.5$, $L_{\max} = 20$ run, taken as a high-fidelity proxy for the true posterior). We report

$$\text{bias} \equiv \max_{d=1,\ldots,21} \frac{|\hat{\mu}_d - \mu^\star_d|}{\sigma^\star_d}, \quad \text{std-ratio} \equiv \Big[\min_d \tfrac{\hat{\sigma}_d}{\sigma^\star_d}, \max_d \tfrac{\hat{\sigma}_d}{\sigma^\star_d} \Big], \tag{1}$$

with bias measured in units of the reference posterior standard deviation per dimension.

Two qualitative observations from this comparison (Table **??**). First, at small production length ($N_{\rm prod} =$ 1,000), unadjusted methods report ESS $\approx$ 20–30 from 64,000 ensemble draws, seemingly an order of magnitude behind Metropolized PICKLES. This is largely a Sokal autocorrelation *truncation* artifact: at small N, the autocorrelation estimator under-counts the long autocorrelation tail and over-states the gap. At long production length the gap shrinks substantially as the autocorrelation tail is properly resolved.

Second, when given a step-size near the BAOAB stability boundary, unadjusted PICKLES can *exceed* Metropolized PICKLES in ESS / grad. At $\eta = 0.10$ ($\gamma = 0.5$, no adapter) we measure ESS/grad $= 1.15\times10^{-2}$, 2.8$\times$ higher than Metropolized PICKLES at the matched $\gamma = 0.5$ tuning, with bias $\leq 0.09\sigma_{\rm ref}$ in posterior mean and a $[0.91, 1.03]$ standard-deviation ratio (within statistical noise of the Metropolized reference). Each unadjusted leapfrog step contributes one sample, while each Metropolized iteration consumes $L = 20$ leapfrog steps to produce one accepted sample; when the BAOAB discretization is naturally accurate at the chosen η, paying for the Metropolis correction costs grad efficiency without buying additional accuracy.

Table 5: Unadjusted-method comparison on DES Y1, long-budget for fair asymptotic comparison; bias measured against the Metropolized PICKLES reference. The Metropolized PICKLES and NUTS-dense rows are reproduced from §?? for context.

	$N_{\rm ch}$	$N_{\rm prod}$	η	ESS	ESS / grad	bias
Metropolized (reference)						
PICKLES ($\gamma = 0.5$, $L_{\max} = 20$)	64	5,000	0.07 (adapt)	26,151	4.09×10^{-3}	0
NUTS-dense	4	5,000	0.13 (adapt)	2,385	3.72×10^{-3}	0
Unadjusted						
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.10	1,477	$\mathbf{1.15\times10^{-2}}$	0.09σ
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.05	270	2.11×10^{-3}	0.04σ
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.20	—	— (NaN)	diverged
PICKLES ($\gamma = 0.5$, BAOAB, adaptive)	32	10,000	0.065 (adapt)	588	1.84×10^{-3}	0.06σ
ALDI (overdamped, PGN-adapted)	64	20,000	0.009 (adapt)	1,543	1.20×10^{-3}	0.04σ

The decisive caveat is that the bias-stability boundary must be respected by hand. At $\eta{=}0.20$, BAOAB diverges to NaN within the warmup phase. ALDI (overdamped) cannot reach the high-η regime at all: without the projected-gradient-norm adapter, ALDI diverges at $\eta \geq 0.05$. The kinetic BAOAB integrator's larger stability margin (via the momentum variable) is what allows unadjusted PICKLES to operate at $\eta = 0.10$. Metropolized PICKLES eliminates this risk: at any step size, the Metropolis-adjusted update is provably unbiased and divergence-free regardless of the acceptance rate. We therefore recommend Metropolized PICKLES as the practical default in cosmology, with unadjusted PICKLES as a higher-efficiency alternative when the discretization bias has been independently verified.

B Per-seed data

The main-text tables report multi-seed medians. This appendix gives the per-seed ESS / grad and acceptance values that underlie the medians. Per-seed scatter is large for some configurations (notably PICKLES $\gamma = 0.5$, $w_s = 0$), and the median is dominated by which 2 of 3 seeds happen to lie in the same basin: this is the "small-sample-size median noise" issue discussed in §??. Cells shown in red are collapsed seeds (ESS / grad $< 10^{-3}$, the bias-mixing bottleneck where the empirical-covariance preconditioner $\hat{\Sigma}$ fails to align with the $\sigma_8 \leftrightarrow b$ ridge).

Table 6: Per-seed ESS / grad (acceptance) at $N_{\rm prod} = 1000$, fid-ball init, $w_s = 0$ for PICKLES variants and emcee, NUTS uses its own dual-averaging warmup. Underlies Table ??.

sampler	seed 0	seed 1	seed 2	median
PICKLES (γ=0.5, $L_{\max}$=20)	5.90×10^{-3} (0.95)	5.56×10^{-3} (0.95)	3.74×10^{-5} (0.56)	5.56×10^{-3} (0.95)
PICKLES (γ=0.5, $L_{\max}$=50)	4.61×10^{-3} (0.93)	5.01×10^{-3} (0.93)	4.74×10^{-5} (0.61)	4.61×10^{-3} (0.93)
PICKLES (γ=0.5, $L_{\max}$=128)	3.60×10^{-3} (0.92)	4.16×10^{-3} (0.92)	8.06×10^{-4} (0.67)	3.60×10^{-3} (0.92)
PICKLES (default γ=2)	3.64×10^{-3} (0.94)	3.98×10^{-3} (0.94)	1.93×10^{-4} (0.99)	3.64×10^{-3} (0.94)
NUTS-dense	7.55×10^{-3} (0.80)	4.31×10^{-3} (0.80)	4.69×10^{-3} (0.80)	4.69×10^{-3} (0.80)
NUTS-block	5.36×10^{-3} (0.80)	4.03×10^{-3} (0.80)	3.32×10^{-3} (0.80)	4.03×10^{-3} (0.80)
NUTS-diag	2.01×10^{-3} (0.80)	2.48×10^{-3} (0.80)	2.27×10^{-3} (0.80)	2.27×10^{-3} (0.80)
`emcee`/stretch	2.87×10^{-4} (0.28)	2.67×10^{-4} (0.27)	2.87×10^{-4} (0.27)	2.87×10^{-4} (0.27)

Per-seed inspection of Table ?? reveals that all PICKLES $\gamma = 0.5$ rows have one collapsed seed (seed 2) and two clean seeds; PICKLES $\gamma = 2$ default also has one collapsed seed (still seed 2). NUTS samplers and `emcee` have no collapsed seeds (the bias-mixing pathology is specific to the ensemble-preconditioner-with-no-stretch-warmup combination).

Table 7: Per-seed ESS / grad (acceptance) for the warmup sweep (Table **??**). All cells use prior-median init. Rows are organized by stretch pre-warmup $w_s \in \{200, 50, 0\}$.

config	seed 0	seed 1	seed 2	median
PICKLES $\gamma = 0.5$, $w_s = 200$*:*				
warmup = 50	5.49×10^{-3} (0.83)	5.82×10^{-4} (0.79)	4.32×10^{-3} (0.77)	4.32×10^{-3} (0.79)
warmup = 100	7.16×10^{-3} (0.87)	6.24×10^{-3} (0.85)	5.39×10^{-3} (0.84)	6.24×10^{-3} (0.85)
warmup = 150	5.90×10^{-3} (0.86)	6.79×10^{-3} (0.87)	5.80×10^{-3} (0.85)	5.90×10^{-3} (0.86)
warmup = 200	4.71×10^{-3} (0.83)	5.16×10^{-3} (0.84)	6.52×10^{-3} (0.87)	5.16×10^{-3} (0.84)
PICKLES $\gamma = 0.5$, $w_s = 50$*:*				
warmup = 50	5.65×10^{-3} (0.76)	2.93×10^{-4} (0.68)	4.25×10^{-3} (0.64)	4.25×10^{-3} (0.68)
warmup = 100	4.82×10^{-3} (0.71)	4.20×10^{-3} (0.67)	7.06×10^{-3} (0.68)	4.82×10^{-3} (0.68)
warmup = 200	5.51×10^{-4} (0.60)	4.15×10^{-3} (0.61)	3.05×10^{-4} (0.59)	5.51×10^{-4} (0.60)
PICKLES $\gamma = 0.5$, $w_s = 0$*:*				
warmup = 50	2.76×10^{-4} (0.91)	6.08×10^{-3} (0.94)	3.26×10^{-5} (0.84)	2.76×10^{-4} (0.91)
warmup = 100	5.14×10^{-3} (0.95)	5.72×10^{-3} (0.94)	3.97×10^{-5} (0.73)	5.14×10^{-3} (0.94)
warmup = 150	6.47×10^{-3} (0.86)	2.88×10^{-4} (0.95)	5.67×10^{-3} (0.94)	5.67×10^{-3} (0.94)
warmup = 200	5.56×10^{-4} (0.96)	5.10×10^{-3} (0.93)	4.86×10^{-5} (0.69)	5.56×10^{-4} (0.93)
PICKLES $\gamma = 2$, $w_s = 0$ *(3-seed at all warmups):*				
warmup = 50	3.42×10^{-3} (0.83)	5.15×10^{-4} (0.95)	3.33×10^{-3} (0.96)	3.33×10^{-3} (0.95)
warmup = 100	4.16×10^{-3} (0.84)	3.72×10^{-3} (0.93)	3.42×10^{-3} (0.90)	3.72×10^{-3} (0.90)
warmup = 150	3.78×10^{-3} (0.80)	3.49×10^{-3} (0.92)	3.93×10^{-3} (0.93)	3.78×10^{-3} (0.92)
warmup = 200	3.17×10^{-3} (0.77)	3.93×10^{-3} (0.94)	3.51×10^{-3} (0.93)	3.51×10^{-3} (0.93)
NUTS-dense (3-seed at all warmups):				
warmup = 50	1.77×10^{-5} (0.60)[a]	2.79×10^{-3} (0.93)[g]	1.42×10^{-3} (0.88)[h]	1.42×10^{-3} (0.88)
warmup = 100	3.36×10^{-3} (0.95)	5.99×10^{-3} (0.89)[d]	3.98×10^{-3} (0.96)	3.98×10^{-3} (0.95)
warmup = 150	7.48×10^{-4} (0.92)	1.58×10^{-3} (0.92)	9.00×10^{-4} (0.90)	9.00×10^{-4} (0.92)
warmup = 200	1.59×10^{-3} (0.89)[b]	2.83×10^{-3} (0.94)[e]	2.04×10^{-3} (0.90)[f]	2.04×10^{-3} (0.90)
NUTS-block (3-seed at all warmups):				
warmup = 50	1.75×10^{-3} (0.77)[c]	1.49×10^{-3} (0.78)[i]	2.47×10^{-3} (0.87)[j]	1.75×10^{-3} (0.78)
warmup = 100	3.30×10^{-3} (0.96)	2.85×10^{-3} (0.96)	2.91×10^{-3} (0.96)	2.91×10^{-3} (0.96)
warmup = 150	2.12×10^{-3} (0.93)	1.64×10^{-3} (0.94)	2.42×10^{-3} (0.94)	2.12×10^{-3} (0.94)
warmup = 200	3.67×10^{-3} (0.95)	2.37×10^{-3} (0.95)	2.10×10^{-3} (0.95)	2.37×10^{-3} (0.95)

[a] 314 divergences (seed 0); mass matrix not converged. [g] 4 divergences (seed 1). [h] 25 divergences (seed 2). [b] 28 divergences (seed 0). [e] 1 divergence (seed 1). [f] 3 divergences (seed 2). [c] 80 divergences (seed 0). [i] 36 divergences (seed 1). [j] 3 divergences (seed 2). [d] 5 divergences.

How does collapse manifest dimensionally? For each collapsed seed (ESS / grad $< 10^{-3}$), the bottleneck dimension — the one whose marginal autocorrelation is the slowest — falls into one of two qualitative patterns:

- *Full-ridge collapse.* All six dimensions of the slow subspace (σ_8, b_1, b_2, b_3, b_4, b_5) simultaneously bottleneck at ess $\approx$ 40–70. This is the dominant collapse mode for PICKLES $\gamma = 0.5$, $w_s = 0$ with random walker positions that fail to capture the $|\rho| \approx 0.94$ correlation ridge in $\hat{\Sigma}$. Example: PICKLES $\gamma = 0.5$, $L_{\max} = 20$, $w_s = 0$, seed 2 has worst-5 dims $\sigma_8(48)$, $b_2(52)$, $b_3(55)$, $b_1(57)$, $b_5(63)$ — the entire slow subspace stalled.
- *Partial-dim collapse.* A single dimension (most often b_1, occasionally Ω_c) is bottlenecked while the other slow modes remain at ESS $\sim$ 5000–8000. This is the typical mode for PICKLES $\gamma = 0.5$, $w_s = 0$ at warmup $= 150, 200$ seed 1 (b_1 at 369, σ_8 at 5758) and warmup $= 200$ seed 0 (Ω_c at 711, others $\gtrsim 4500$). Mechanism: $\hat{\Sigma}$ captured most of the ridge but failed to align one specific eigendirection; the chain mixes well in 5 of the 6 slow modes but stalls in the orthogonal 6th.

For NUTS at low warmup, the failure is qualitatively different — it is integrator instability (a non-zero divergence count signals that the dual-averaged mass matrix has not converged), not preconditioner mis-

alignment. Bottleneck dimensions are spread roughly uniformly across the slow subspace; NUTS does not have the random-walker-init $\hat{\Sigma}$ pathology because its mass matrix is dual-averaged from its own iterations rather than estimated from a 64-walker ensemble.

Table 8: Bottleneck dimension for each collapsed seed (ESS / grad $< 10^{-3}$). "Pattern" classifies the dimensional structure of the collapse: full-ridge (all 6 slow dims bottlenecked together), partial-dim (one dim collapsed, rest of slow subspace at typical ESS), or NUTS-divergence (low ESS from integrator instability). "worst dim" is the marginal with the lowest Sokal ESS.

config (seed)	worst dim	ess_{min}	2nd-worst	pattern
PICKLES γ=0.5 $L_{\max}$=20 w_s=0 w= 500 (s2)	σ_8	48	b_2(52)	full-ridge
PICKLES γ=0.5 $L_{\max}$=50 w_s=0 w= 500 (s2)	b_5	152	b_2(235)	full-ridge
PICKLES γ=0.5 $L_{\max}$=128 w_s=0 w= 500 (s2)	b_3	6,591	b_1(6,613)	full-ridge (weak)
PICKLES γ=2 default w_s=0 w= 500 (s2)	σ_8	247	b_1(370)	full-ridge
PICKLES γ=0.5 w_s=200 w= 50 (s1)	σ_8	745	b_1(6,665)	partial-dim
PICKLES γ=0.5 w_s=50 w= 50 (s1)	σ_8	374	b_1(722)	full-ridge (weak)
PICKLES γ=0.5 w_s=50 w= 200 (s0)	σ_8	705	b_1(720)	full-ridge (weak)
PICKLES γ=0.5 w_s=50 w= 200 (s2)	σ_8	390	b_1(716)	full-ridge (weak)
PICKLES γ=0.5 w_s=0 w= 50 (s0)	Ω_c	353	b_4(385)	broad (incl. cosmo)
PICKLES γ=0.5 w_s=0 w= 50 (s2)	b_5	42	b_3(45)	full-ridge
PICKLES γ=0.5 w_s=0 w= 100 (s2)	b_3	51	σ_8(67)	full-ridge
PICKLES γ=0.5 w_s=0 w= 150 (s1)	b_1	369	σ_8(5,758)	partial-dim
PICKLES γ=0.5 w_s=0 w= 200 (s0)	Ω_c	711	σ_8(4,577)	broad (incl. cosmo)
PICKLES γ=0.5 w_s=0 w= 200 (s2)	σ_8	62	b_2(64)	full-ridge
NUTS-dense warmup = 50 (s0)	—	3	—	NUTS divergence (314)
NUTS-dense warmup = 150 (s0)	—	391	—	NUTS divergence (3)
NUTS-dense warmup = 150 (s2)	—	450	—	NUTS divergence (10)
NUTS-dense warmup = 200 (s0)	—	320	—	NUTS divergence (28)
NUTS-block warmup = 50 (s0)	—	412	—	NUTS divergence (80)

The collapse-pattern classification of Table **??** makes the structural-vs-stochastic distinction explicit. For $w_s = 200$ PICKLES, no collapses occur. For $w_s = 0$, the dominant collapse mode is full-ridge (PICKLES $\gamma = 0.5$, seeds with inadvertent walker positions that don't span the $\sigma_8 \leftrightarrow b$ correlation), with occasional partial-dim collapses where the ensemble preconditioner captured 5 of 6 ridge eigendirections but missed one.

The per-seed scatter in Table **??** makes two points concrete. First, for PICKLES $\gamma = 0.5$, $w_s = 0$, the collapse rate is approximately stochastic across warmup values (2/3, 1/3, 1/3, 2/3 at warmup $= 50, 100, 150, 200$); the median shuffles between collapsed and good basins depending on the random seed assignment, so the apparent non-monotonicity in warmup is in large part a 3-seed median artifact. Second, increasing w_s from 0 to 200 unambiguously reduces collapse rate from 1–2 of 3 seeds down to 0 of 3 seeds, confirming that stretch pre-warmup is the genuine fix.

C Configuration summary for reproducibility

Common setup. Synthetic DES Y1 3×2pt mock data generated from the `jax-cosmo` forward model at the fiducial ΛCDM cosmology of Campagne et al. [**?**]; data vector $\in \mathbb{R}^{2250}$, Gaussian likelihood with the Campagne+ 3×2pt covariance. Posterior dimension $D = 21$ (six cosmology + two intrinsic-alignment + five galaxy bias + four shear-calibration + four photo-z shift parameters). All samplers run in the unconstrained z-space obtained by logit transform on the 13 Uniform priors and non-centered (location-scale) reparametrization on the 8 Gaussian priors, so the prior is $\mathcal{N}(0, I_{21})$ in z-space. Hardware: single NVIDIA H100 80GB GPU. Software: JAX 0.9.2, NumPyro 0.20.1, `jax-cosmo` 0.1.0, NumPy 2.4.4. ESS is computed from the integrated autocorrelation time using the monotone-positive Sokal estimator, reported here *without* the $\tau \geq 1$ clamp (so antithetic NUTS chains can report ESS $> N$); the worst dimension over $D = 21$ is reported as ESS_{min}. Gradient counts are production-only (warmup gradients excluded).

Per-table experimental settings.

- Table **??** (main comparison, $N_{\text{prod}} = 1000$): 3 seeds, median reported. Walkers initialized $z \sim \mathcal{N}(z_{\text{fid}}, 0.5^2 I)$ (fid-ball init). Gradient-sampler warmup = 500 iterations. No stretch pre-warmup ($w_s = 0$). PICKLES variants: $N_{\text{ch}} = 64$ (`jax.vmap` parallel), $L = L_{\max} \in \{20, 50, 128\}$, $\gamma \in \{0.5, 2\}$. `emcee`/stretch: $N_{\text{ch}} = 64$ (same ensemble configuration as PICKLES); stretch parameter $a = 2$. NUTS variants (`numpyro`): $N_{\text{ch}} = 4$ run sequentially (`chain_method = "sequential"`), max-tree-depth = 7, target-accept = 0.8, `init_to_median`.
- Table **??** (long-production reference, $N_{\text{prod}} = 5000$): single seed, same per-sampler configuration as Table **??**; `emcee` uses $N_{\text{prod}} = 50{,}000$ to resolve its long autocorrelation tail.
- Table **??** (per-parameter breakdown, $N_{\text{prod}} = 1000$): 3 seeds, median reported. PICKLES uses the *prior-median* initialization $z \sim \mathcal{N}(0, 0.5^2 I)$ (the NUTS-fair center) with $w_s = 200$ iterations of stretch pre-warmup [**?**], then 200 gradient-sampler warmup and 1000 production samples at $\gamma = 0.5$, $L_{\max} = 20$, $N_{\text{ch}} = 64$. NUTS-dense as in Table **??** ($N_{\text{ch}} = 4$, full 21×21 mass matrix, max-tree-depth 7).
- Table **??** (warmup-length sweep, $N_{\text{prod}} = 1000$): 3 seeds, median reported. Walkers prior-median initialized ($z_{\text{init}} \sim \mathcal{N}(0, 0.5^2 I)$). PICKLES variants reported under three stretch pre-warmup choices $w_s \in \{0, 50, 200\}$ and two damping choices $\gamma \in \{0.5, 2\}$ at $L_{\max} = 20$. NUTS variants under their own dual-averaging warmup (no stretch pre-warmup applies). Gradient-sampler warmup is varied across $\{50, 100, 200\}$ iterations.
- Table **??** (unadjusted-method appendix): single seed. Unadjusted PICKLES uses BAOAB integration without Metropolis correction. $N_{\text{ch}} = 32$ or 64 as noted in the N_{ch} column; fixed step size η as noted ($\eta = 0.05,\ 0.10,\ 0.20$); fid-ball init ($z \sim \mathcal{N}(z_{\text{fid}}, 0.5^2 I)$), no stretch pre-warmup, 200-sample burn-in for bias measurement. ALDI uses the PGN-adapted variant with $\eta_{\text{init}} = 0.009$; $N_{\text{prod}} = 20{,}000$ at $N_{\text{ch}} = 64$.

Diagnostics. *Acceptance* is the time-averaged Metropolis acceptance rate over all production iterations and chains; for NUTS the `mean_accept_prob` field of the post-warmup adapter. *Divergences*: the count of production iterations on which the NUTS leapfrog energy-error exceeded the `numpyro` default threshold ($1000\, k_B T$ in the Hamiltonian analogy; configurable via `max_delta_energy`); a non-zero divergence count indicates the integrator is unstable along part of the typical set, and is the standard signal that the dual-averaged mass matrix or step size is mistuned. $\bar{L}$ is the mean number of leapfrog steps per accepted sample (fixed for PICKLES at $L_{\max}$; adaptive for NUTS via the no-U-turn criterion).

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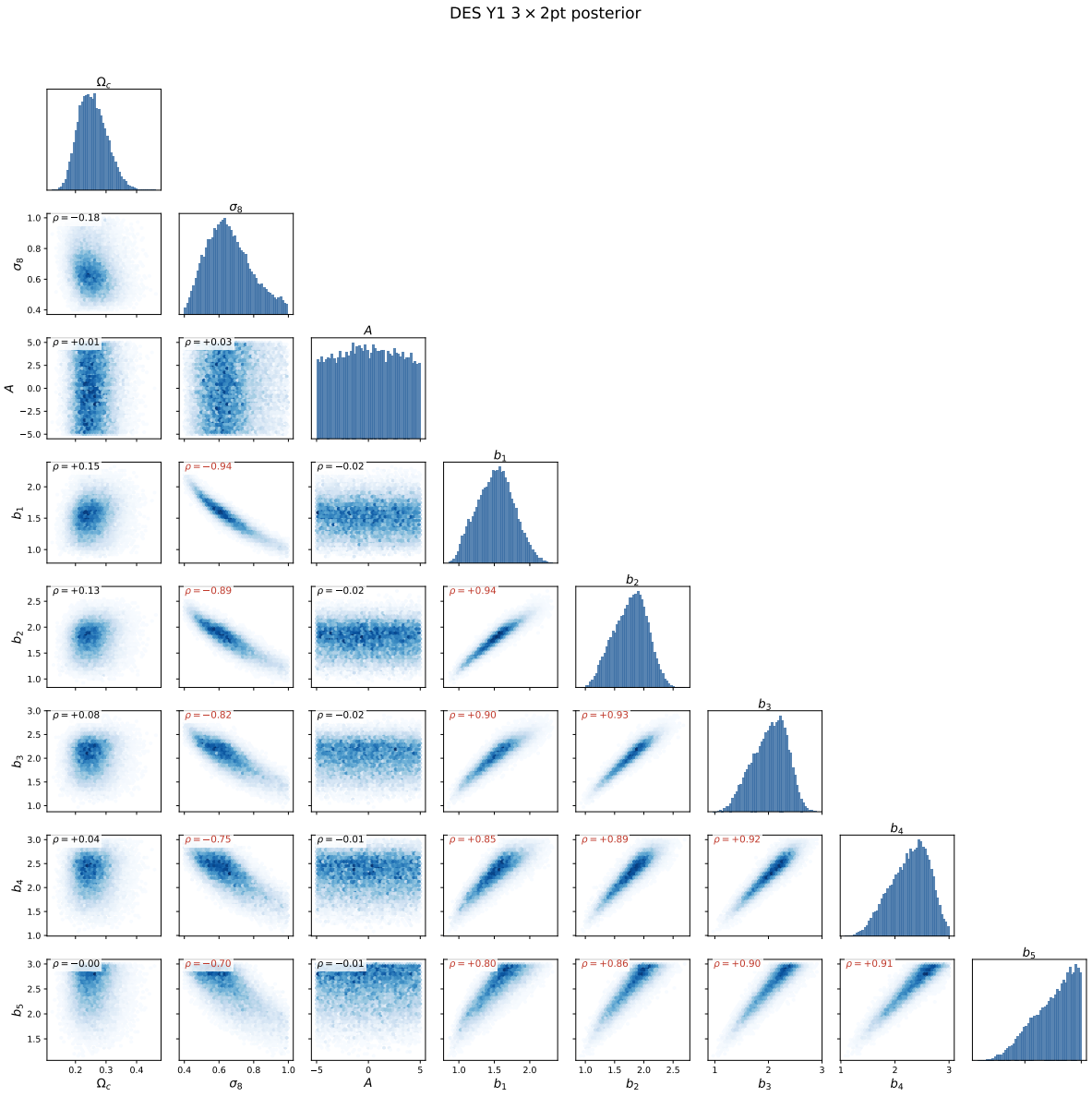


Figure 1: DES Y1 3 × 2pt posterior, eight *sampled* marginals of $D = 21$. The two control panels (rows Ω_c, A) show near-isotropic ellipses with $|\rho| \leq 0.2$, while every panel involving σ_8 and the five galaxy biases $b_{1:5}$ shows a narrow ridge ($|\rho|$ from 0.70 to 0.94, red text). The slow-mixing 6-dim subspace consists entirely of $\sigma_8 + b_{1:5}$ (§**??**); the remaining 13 dimensions (Ω_b, h, n_s, w_0, η, $m_{1:4}$, $\Delta z_{1:4}$) mix to the trivial upper bound and are not shown.

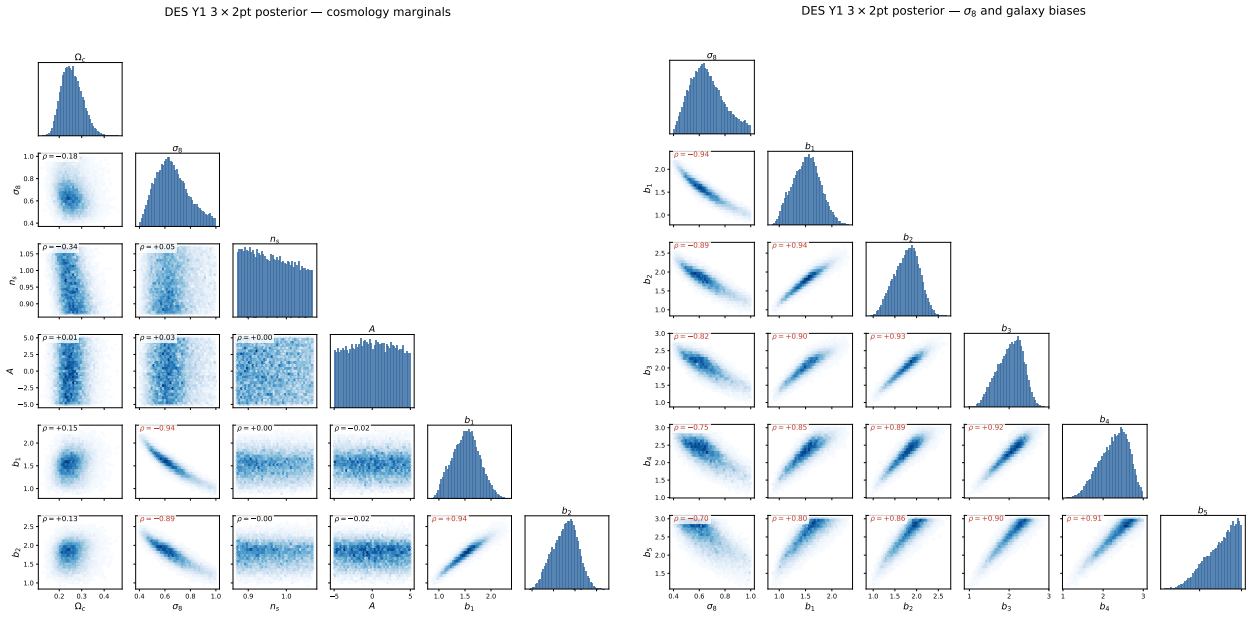


Figure 2: Alternative six-dimensional views of the posterior. Left: the conventional cosmology-style corner (Ω_c, σ_8, n_s, A, b_1, b_2) showing three ridges ($\sigma_8 \leftrightarrow b_1$, $\sigma_8 \leftrightarrow b_2$, $b_1 \leftrightarrow b_2$, each at $|\rho| \approx 0.94$). Right: the slow subspace (σ_8 + the five galaxy biases $b_{1:5}$), in which every off-diagonal panel is a narrow ridge ($|\rho|$ from 0.70 to 0.94); this is the 6-dim subspace responsible for the ESS bottleneck in Table **??**.